Project Development Phase

Electric Vehicle Market Performance & Insights

This phase records how the planned EV market analysis was implemented in Tableau Public, from connecting the prepared dataset to assembling the worksheets, dashboard, and final Story. It focuses on the implementation workflow rather than repeating the complete design specification.

1. Development Objective

The objective was to turn the prepared electric-vehicle dataset into an interactive, readable Tableau analysis that helps users explore sales, manufacturers, regions, vehicle categories, battery characteristics, range, and price distribution. The Story was arranged as a guided sequence and ends with a text-based dashboard containing key insights, strategic recommendations, and a conclusion.

2. Development Environment and Input

Visualization tool	Tableau Public
Primary input	ev_market_preprocessed_combined.csv
Supporting data-quality file	ev_market_data_quality_report.csv
Supporting notes	ev_market_preprocessing_notes.txt
Prepared dataset size	299,999 rows × 38 columns
Data preparation summary	Train and test files combined; one exact duplicate removed; record_source field added; text whitespace trimmed; blank text converted to missing; 28,507 missing cells retained for transparent handling.

3. Implementation Workflow

1. **Connect the prepared data:** Open Tableau Public and connect to ev_market_preprocessed_combined.csv. Confirm that the expected fields are available and that the source loads successfully.
2. **Review field roles and data types:** Check that numeric measures such as annual sales, price, battery capacity, and range are interpreted as numeric fields; check that geographic and category fields have suitable roles.
3. **Create the worksheets:** Build the planned views for sales trend, top manufacturers, region versus category, manufacturer treemap, category-wise sales, EV distribution map, capacity versus range, battery technology comparison, and price distribution.
4. **Configure measures and chart logic:** Use appropriate aggregations for each view. For example, use SUM for annual sales, AVG for numeric characteristics where appropriate, and distinct counting for vehicle identifiers. Confirm the exact field names and units in the connected dataset before finalizing calculations.
5. **Assemble the main dashboard:** Place the relevant visualizations and KPI cards into the main dashboard, keeping labels, legends, and layout readable. Use consistent formatting and avoid implying that vehicle IDs necessarily represent commercial model names.
6. **Build the Tableau Story:** Add the dashboard and worksheet views in the intended sequence. Set each Story view to Entire View so it fits within the Story Point area.
7. **Add the final synthesis dashboard:** After Price Distribution, add the final dashboard containing key insights, strategic recommendations, and conclusion text. Ensure the written takeaways reflect patterns actually observed in the charts.
8. **Review and save the workbook:** Check titles, captions, tooltips, filters, legends, number formats, and the order of Story points. Save the workbook and publish to Tableau Public if required for submission or portfolio sharing.

4. Worksheets Implemented

The planned visual analysis consists of the following worksheet views. This list records the implemented view set; the detailed chart rationale and dashboard layout are documented in the Project Design Phase.

Worksheet	Implementation purpose
Sales Trend	Line chart showing change in annual sales across the available time field.
Top 10 Manufacturers	Ranked bar chart comparing the leading manufacturers by the selected sales measure.
Region vs Category	Heatmap or highlight table comparing vehicle categories across regions.
Manufacturer Treemap	Treemap showing the relative contribution of manufacturers by the selected measure.
Category-wise Sale	Donut chart showing the share of sales across categories.
EV Distribution Map	Geographic map showing distribution where valid geographic data is available.
Capacity vs Range	Scatter plot exploring the relationship between battery capacity and driving range.
Battery Tech Comp	Pie chart comparing the available battery-technology categories.
Price Distribution	Box plot showing price spread and potential outliers.

5. Final Story Structure

9. Dashboard
10. Sales Trend
11. Top 10 Manufacturers
12. Region vs Category
13. Manufacturer Treemap
14. Category-wise Sale
15. EV Distribution Map
16. Capacity vs Range
17. Battery Tech Comp
18. Price Distribution
19. Key Insights, Strategic Recommendations & Conclusion Dashboard

6. Development Decisions and Data Handling

- The combined preprocessed CSV is used as the common source so that worksheet comparisons are based on the same prepared dataset.
- Missing values were retained during preprocessing rather than automatically imputed. In Tableau, fields with missing values should be checked before drawing conclusions from averages, counts, maps, or comparisons.
- The record_source field can support source-level checks where relevant, but it should only be used in analysis when it answers a defined question.
- KPI labels should clearly state the measure and unit. Average price should be labelled in USD if the source field is price_usd; driving range should be labelled in kilometres if the field is range_km.
- If a distinct vehicle_id count is used, label it as distinct vehicle IDs or vehicle records rather than claiming it is a count of commercial EV models unless the data dictionary confirms that meaning.
- Recommendations on the final text dashboard should be written only after reviewing the actual displayed results; do not insert unsupported rankings, percentages, trend directions, or correlation claims.

7. Development Deliverables

- Tableau workbook containing the completed worksheets and dashboard(s).
- Tableau Story containing the nine analytical worksheet views plus the main dashboard and final insight/recommendation dashboard, in the agreed sequence.
- Final text dashboard summarizing evidence-based insights, strategic recommendations, and conclusion.
- Prepared CSV and supporting data-quality/preprocessing notes used during development.

8. Completion Checklist

☐ Prepared dataset connects and all expected fields are available.

☐ All planned worksheets open and display correctly.

☐ KPI calculations, aggregation levels, units, and labels have been checked.

☐ Dashboard layout, legends, tooltips, and any filters have been reviewed.

☐ All Story points are in the intended order and set to Entire View.

☐ Final insight dashboard follows Price Distribution and uses findings supported by the charts.

☐ Workbook is saved and, if required, published with a working share link.

Status note: This document describes the implementation workflow and expected deliverables. It does not claim that publishing, every interaction, or every test has passed; those outcomes should be recorded after checking the actual Tableau workbook.